Quiz 2

Name: _____ Student ID: _____ Marks: 20

Note: This quiz contains four questions with 20 marks in total. Please answer all questions. Do not round off your answer. You must show all necessary steps to receive full mark.

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(1) Evaluate the line integral $\int_C (x + 2y)\, dx + x^2 dy$, where C consists of line segments from $(0, 0)$ to $(2, 1)$ and from $(2, 1)$ to $(3, 0)$. [5]

(2) Find the work done by the force field $\mathbf{F}(x, y) = x^2\ \mathbf{i} + xy\ \mathbf{j}$ on a particle that moves once around the circle $x^2 + y^2 = 4$ oriented in the counterclockwise direction. [5]

(3) Let $\mathbf{F} = 2x\mathbf{i} + \mathbf{j} + 2z\mathbf{k}$. Determine whether or not $\mathbf{F}$ is a conservative vector field. If $\mathbf{F}$ is conservative, find a function f such that $\mathbf{F} = \nabla f$. [5]

(4) Given $\mathbf{F} = \frac{x}{\sqrt{x^2+y^2+z^2}}\ \mathbf{i} + \frac{y}{\sqrt{x^2+y^2+z^2}}\ \mathbf{j} + \frac{z}{\sqrt{x^2+y^2+z^2}}\ \mathbf{k}$. Find curl $\mathbf{F}$ and div $\mathbf{F}$. [5]